House Price Prediction Using Machine Learning

1. Title page

- **Project Title:** House Price Prediction System Using Machine Learning (Linear Regression)
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- **Company \ Institution Name:** Fortune IT Pvt Ltd
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2. Introduction

The real estate industry relies heavily on accurate property valuation. Determining the correct price of a house manually can be difficult because house prices depend on multiple factors such as location, size, number of rooms, and age of the property.

This project aims to build a **Machine Learning model using Linear Regression** that can predict house prices based on historical housing data.

The system helps **buyers, sellers, and real estate agents** estimate property prices quickly and accurately.

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3. Objective of the Project

The main objective of this project is to develop a **predictive model** that estimates house prices based on various property features.

The system will:

- Analyze housing data
- Identify relationships between property features and price
- Predict the estimated price of a house

This helps in **data-driven decision making in the real estate market**.

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4. Problem Statement

House prices vary depending on many factors such as:

- Area of the house
- Number of bedrooms

- Location
- Age of the property

Manually predicting house prices can lead to inaccurate estimates.

Therefore, this project uses **Machine Learning with Linear Regression** to automatically analyse data and predict house prices more accurately.

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5. Dataset Description

The dataset used in this project contains historical housing information.

Dataset Features

Feature	Description
Area	Size of the house in square feet
Bedrooms	Number of bedrooms in the house
Location	Area or city where the house is located
Age	Age of the house in years
Price	Actual house price (Target variable)

The **Price column is the target variable**, which the model will learn to predict.

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6. Tools and Technologies Used

Programming Language

- Python

Libraries

- **Pandas** – Data analysis and manipulation
- **NumPy** – Numerical computations
- **Scikit-learn** – Machine learning model implementation
- **Matplotlib** – Data visualization
- **Seaborn** – Statistical visualization

Development Environment

- Jupyter Notebook
  or
- Visual Studio Code (VS Code)

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7. Machine Learning Algorithm Used

Linear Regression

Linear Regression is a **supervised machine learning algorithm** used for predicting continuous values.

It identifies the relationship between **independent variables (features)** and the **dependent variable (house price)**.

The model learns from historical data and predicts future house prices.

Example formula:

Price = m1 × Area + m2 × Bedrooms + m3 × Age + b

Where:

- m = coefficients learned by the model
- b = intercept

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8. Project Workflow

The project follows a structured machine learning pipeline.

Step 1: Import Libraries

Import required Python libraries such as Pandas, NumPy, and Scikit-learn.

Step 2: Load Dataset

Load the housing dataset into the Python environment using Pandas.

Step 3: Data Preprocessing

Data cleaning is performed to prepare the dataset for machine learning.

Tasks include:

- Handling missing values
- Removing duplicate data
- Encoding categorical variables like location

Step 4: Exploratory Data Analysis (EDA)

EDA is performed to understand the dataset.

This includes:

- Checking data distribution
- Visualizing relationships between variables
- Identifying correlations

Step 5: Train-Test Split

The dataset is divided into two parts:

- **Training Data (80%)** – Used to train the model
- **Testing Data (20%)** – Used to evaluate the model

Step 6: Model Training

The Linear Regression model is trained using the training dataset.

Step 7: Prediction

After training, the model predicts house prices for new data.

Step 8: Model Evaluation

The model performance is evaluated using standard metrics.

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9. Model Evaluation Metrics

To measure model performance, the following metrics are used:

Mean Absolute Error (MAE)

Measures the average absolute difference between predicted and actual prices.

Mean Squared Error (MSE)

Measures the average squared difference between predicted and actual values.

R^2 Score

Indicates how well the model explains the variance in the data.

Higher R^2 score indicates better model performance.

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10. Expected Output

Example input to the model:

Area = 2000 sq ft
Bedrooms = 3
Location = City Center
Age = 5 years

Predicted Output

Estimated House Price = **₹85,00,000**

The system will automatically calculate the predicted price based on the trained model.

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11. Applications of the Project

This project has real-world applications in multiple domains.

Real Estate Industry

Helps property dealers estimate house prices.

Property Valuation Systems

Automated systems for calculating property value.

Housing Market Analysis

Used by analysts to study housing market trends.

Online Property Platforms

Websites can use this model to suggest house prices.

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12. Conclusion

This project demonstrates how **Machine Learning can be applied to real estate price prediction**.

Using **Linear Regression**, the model analyses historical housing data and predicts house prices based on property features.

The system improves **accuracy, efficiency, and decision-making** in the real estate industry.

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13. Future Enhancements

The project can be improved further by:

- Using larger real estate datasets
- Applying advanced algorithms like:
    - Random Forest
    - Gradient Boosting
    - XGBoost
- Building a **web application using Flask or Streamlit**
- Deploying the model for real-time predictions